PY-STEM · ACTIVITY PYS-12 · DESIGN

Accessibility Ramp Rescue Lab

Design a portable ramp that works for a real user constraint.

Win condition: Best safe ramp meeting slope, load, portability, and client criteria.

THE SCIENCE YOU ARE USING

Slope, load, and universal design. A ramp trades steepness for length: a gentler slope is easier and safer to use but needs more room. Accessibility standards cap how steep a ramp can be. Universal design means building for real users and real constraints from the start, not as an afterthought.

Criteria and constraints. Your client gives criteria: a maximum slope, a load to carry, and a portability limit. You build a scale ramp for a toy wheelchair or weighted cart that meets all of them, then defend the tradeoffs.

HOW TO BUILD AND RUN IT

01 **Read the client criteria.** Note the maximum slope, the required load, and the portability limit.

02 **Sketch to meet the slope.** Work out the ramp length that keeps the slope within the limit for the given height.

03 **Build the scale ramp.** Construct a sturdy ramp from foam board and supports that folds or carries easily.

04 **Load test.** Roll the weighted cart up and down. Confirm it holds the load without sagging.

05 **Defend the tradeoffs.** Explain how your design meets slope, load, and portability together.

Safety: No standing on ramps. Loads are small tabletop weights only. Allergy: None.

MATERIALS

Foam board	per team
Craft sticks	shared
Dowels	shared
Binder clips	shared
Rulers	per team
Weighted cart	shared
Tape	shared

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): ______________________________

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The ONE variable we are testing: ______________________________

Baseline result: ______________________________

After redesign: ______________________________

DEFEND YOUR DESIGN

What evidence made you change your design? ______________________________

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If you ran it again, what is the ONE thing you would change? ______________________________

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HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Load held	30
Slope meets target	30
Portability	20
Client criteria explanation	20
Total	**100**